

Principles of Orthopaedic Rehabilitation

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The purpose of this chapter is to describe the principles of orthopaedic rehabilitation. Orthopaedic rehabilitation should balance the load on tissues to stimulate adaptation but not so much to disrupt the healing process. This chapter will describe how loading models can be applied to the rehabilitation process and contrast how this loading is understood to effect different tissues. This paradigm will then be used to identify the key components to designing optimal rehabilitation protocols and pathways while differentiating key interventions and criteria in the context of important patient modifiers. Upon completion of this chapter, orthopaedic specialists will be equipped to develop and implement nonoperative and postoperative protocols or pathways that appropriately balance load and maximize the potential to return patients to their optimal level of function. This chapter will provide an overview of the key concepts and drivers of orthopaedic rehabilitation. The chapter will emphasize the importance of appropriate tissue loading throughout the rehabilitation process.

INTRODUCTION

Envelope of Function

Tissue response to loading has been described as the result of the magnitude and frequency of loading.¹ When load and frequency are matched appropriately, the individual is functioning in his or her “zone of homeostasis,” which is loading that does not result in tissue adaptation or tissue failure (Fig. 31.1). When the magnitude or frequency of loading exceeds the tissue’s physiologic tolerance, the individual begins to function outside of his or her “envelope of function.” This results in supraphysiologic (i.e., stress that is greater than the normal stress of a given person) overload that is positive when the tissues can adapt. For example, when muscle experiences supraphysiologic load over time this results in hypertrophy and neural adaptation. However, when the load exceeds the maximum tissue tolerance threshold, either in magnitude or frequency, the result is structural failure. Following an injury or surgery, tissues are injured as depicted in the figure. This means that activities that were once within the envelope of function are now shifted outside of it. For example, prior to an anterior cruciate ligament (ACL) injury and reconstruction, walking 2 miles was likely within a patient’s zone of homeostasis. However, in the weeks immediately following injury and surgery, this task would likely fall outside of the envelope of function. Thus at the beginning of a rehabilitation period,

the tissues involved should be considered when establishing the loading and motion limitations to set the boundaries of the rehabilitation.

Acute: Chronic Workload Ratio

Workload represents all of the stress a given structure experiences (Fig. 31.2). For example, femoral articular cartilage following a “bone bruise” would undergo stress not only from the rehabilitation exercises but also normal activities of daily living such as walking, going up/down stairs, and squatting to sit down/get up from a chair. Acute workload is most often defined as the workload over the 7 days, and chronic workload over the past 28 days. The relationship between the acute (what is currently imparted on tissues) versus the chronic workload (what stress the tissue has been able to adapt to) plays a critical role in the appropriate progression of individuals as they return to sport.^{2,3} Injury risk has been shown to be a function of the acute to chronic workload ratio, whereby higher ratios increase the likelihood of injury. Simply, acute dramatic increases in load compared with what the body is conditioned to respond to appears to increase injury risk.

The acute load is the load the individual has had over the past week, whereas the chronic load is the load the individual has had over the past 4 weeks. A load that had maintained a steady increase of 10% to 15% per week over time results in a “sweet spot” of the acute to chronic load ratio of approximately 1.3, which is an overall 30% greater load in the past 7 days compared with the past 28 days. In contrast, a recent spike in load, such as an increase of 30% in 1 week (increasing from 45 minutes running to 90 minutes, would result in an acute to chronic workload ratio of 2.0, which has been shown to increase an individual’s risk of injury, which increases rapidly as the ratio increases.

This is important to consider in the rehabilitation process because workload should be increased gradually to allow tissue adaptation and decrease risk of injury. Throughout the rehabilitation process, this risk of injury can also be equated to risk of symptoms. If a patient has an acute spike in loading during rehabilitation, he or she may not have an injury but may return to the clinic with an increase in pain and swelling or a decrease in range of motion (ROM). These symptoms can be predictable based on which tissues are stressed.

Often athletes are injured in the first few months after return to sport, which may be due in part to too large a spike in acute workload relative to their chronic workload during the

Abstract

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Keywords

protocols
loading
rehabilitation progression
tissue healing
range of motion

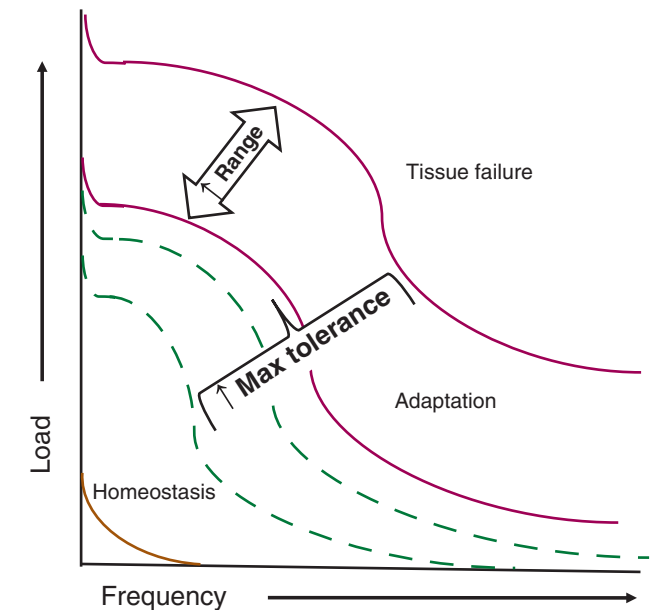


Fig. 31.1 Rehabilitation programs should increase tissue tolerance to the magnitude and frequency of loading, as well as increase the adaptation range of that tissue.

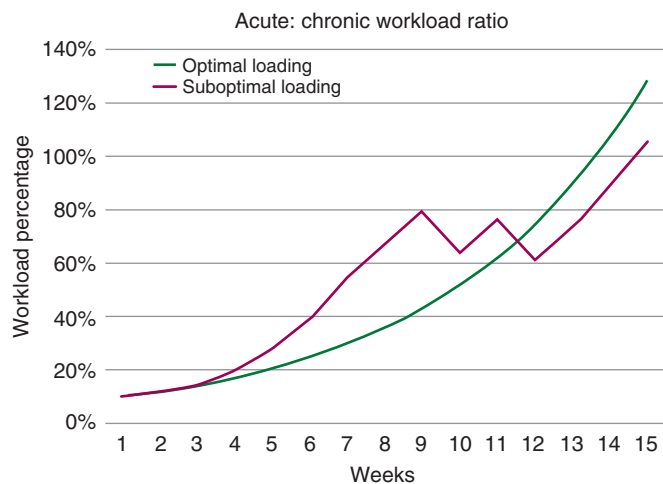


Fig. 31.2 Optimal loading results in the quickest path to return to prior level of function as a period of rest and/or decreased loading is not required to reach loading levels required to return to function.

rehabilitation process. Optimal rehabilitation should progress load gradually from surgery through to full return to activity, with coordination between all providers.

Matched Dosing of Internal and External Load

With those models illustrated, it is important to consider how to apply load, how tissue responds, and how these changes lead to clinical decision-making. External load is what is typically thought of in rehabilitation, encompassed by the sets, reps, and resistance people use to perform an exercise or daily activities. Internal load is the tissue’s response to an external load and can be viewed based on symptoms, including pain, swelling, change in ROM, etc. Matched dosing is a concept that is critical to successful rehabilitation, meaning that the external load should be matched based on the internal load with which a patient presents.

0	Extremely easy
1	
2	Easy
3	
4	Somewhat easy
5	
6	Somewhat hard
7	
8	Hard
9	
10	Extremely Hard

Fig. 31.3 The OMNI rating of perceived exertion provides a meaningful way to document and manage response to a training load. (From Colado JC, Garcia-Masso X, Triplett NT, et al. Construct and concurrent validation of a new resistance intensity scale for exercise with thera-band(R) elastic bands. *J Sports Sci Med.* 2014;13[4]:758-766.)

External load should be increased systematically and gradually, based on bath knowledge of tissue healing, as well as tissue response to loading. If a patient arrives to a rehabilitation session having lost 5 degrees of knee extension ROM with an increase in pain and swelling after jogging last session, this may suggest the internal load is too high, with tissues giving clean warning signs of external overload. This patient’s external load should be decreased until the symptoms of internal load decrease. Clinically, simple methods to measure internal load of the patient include pain, ROM, and rating of perceived exertion (RPE) (Fig. 31.3). The Borg RPE scales are the most commonly used and have been shown to reflect heart rate, breathlessness, and muscle fatigue following exercise.⁴⁻⁶

Tissue Healing Considerations

To prescribe the proper loading plan, understanding the basic tissue biomechanical properties is at the core of forming an effective plan of care in rehabilitation, whether nonoperative, preoperative, or postoperative. Tissues differ in both their ability to heal following injury and in their response to loading. A rehabilitation provider is equipped with the knowledge of these healing capacities and loading responses. This allows them to both unload tissue to the degree required to allow maximal healing while minimizing deleterious effects as well as load the tissue to the degree required for optimal return of function. An understanding of healing capacity and loading response dictates the degree to which specific tissues should be subject to forces across the continuum of loading based on both injury and surgery. Proper communication between a surgeon and rehabilitation provider regarding tissue pathology and surgical procedure allows the rehabilitation provider to incorporate this information into a patient’s plan of care so they are loaded and progressed appropriately at an adaptive level through their course of rehabilitation.

Tissue healing is an important consideration in the development of time-based guidelines in a protocol. Some tissues affected during surgery require a period of immobilization for healing, whereas others can withstand immediate ROM. Likewise, some tissues benefit from limited or no weight bearing for a period of time, whereas others both withstand and benefit from immediate full weight bearing. There are additional patient-specific factors that may influence the timeline for postoperative

restrictions, some of which are patient health, tissue quality, comorbidities, etc. A rehabilitation provider will expect instruction from a patient's surgeon regarding any limitations he or she may have postoperative for tissue healing, including the amount and time frame, regarding weight bearing, immobilization and ROM, and muscle activation. This information should be included in an established postoperative protocol, as well as the patient's postoperative referral to a rehabilitation provider if any modifications from the established protocol are made.

Key Tissue-Specific Considerations

Tendon

Tendon repairs generally require a period of immobilization in a shortened position followed by a gradual increase in ROM and load across the tendon. Healthy tendon responds well to linear loads and eccentric exercises when ready.^{7,8} However, damaged or less healthy tendons (degenerative rotator cuff or Achilles) may not respond as well as acute tendon injuries and are impacted by overall health such as smoking and hypercholesterolemia.⁹⁻¹³

Ligament

Initially, ligamentous reconstructions are typically weakest at the interface between the bone and fixation device, thus restrictions should be based on the strength here. As the ligament graft undergoes "ligamentization," there is a time during which the cellular structure in the graft is disorganized and weaker, and restrictions on activity should be based on the strength of the graft.^{14,15} In contrast, ligamentous repairs are weakest at the site of the repair, which is more tenuous, dictating that immediate postoperative care may be more conservative with a period of immobilization.^{16,17}

Labral/Meniscal

Labral and meniscal tissue can vary in their blood supply based on location of injury and thus can vary in their ability to heal. In general, more vascular areas are more likely to heal, whereas areas with poor blood supply are limited in their capacity to heal.¹⁸ This variance in ability to heal often dictates surgical procedure. In a labral or meniscal débridement, there is no structural healing of repaired tissue postoperatively because the tissue was débrided. As such, postoperative restrictions would be based around the remaining tissue's tolerance to loading. In the lower extremity, the forces distributed across articular cartilage will increase, and loading should be progressive to allow tissue to adapt to this new demand. In a labral or meniscal repair, the tissue should be protected from loading initially with the goal of allowing structural healing to take place to the extent possible while protecting the interface of any fixation. This tissue responds well to compression but not to shear forces, so typical postoperative protocols allow some compressive forces while minimizing shear.

Articular Cartilage

Articular cartilage is also limited in its ability to heal, in part due to lack of blood supply here.¹⁹ Surgical procedures to improve

articular cartilage are either aimed at stimulating a healing response or through transplanting other cartilage to the area. Due to this, there is typically a prolonged period of non-weight bearing if the lesion is on a weight-bearing surface. However, recent animal models show gradual loading and motion aid in healing.²⁰⁻²² This poses a critical challenge for the surgeon and rehabilitation team in designing effective rehabilitation programs following articular cartilage injury. Surgeons may also impose restrictions regarding the degree of motion the joint is allowed based on where in the ROM the lesion would engage.

Bone

Bone responds well to compressive loads and is the most predictable, and radiographs can be used to easily monitor healing.^{23,24} Injuries due to repetitive stress (fifth metatarsal fractures) are not as predictable and should receive special consideration. However, generally 6 weeks of immobilization provides adequate healing time to begin loading.

Patient Modifiers

There are many additional patient-specific factors that put into context tissue considerations. These include established risk factors for delayed healing, such as smoking, diabetes, obesity, nutritional status, and current exercise tolerance.^{25,26} If a patient possesses some of these risk factors, a surgeon may dictate that postoperative rehabilitation progresses at a delayed pace relative to a normal protocol. For example, diabetes is a well-established risk factor for postoperative stiffness in shoulder and knee surgery, likely requiring a modification of the typical postoperative plan.²⁷⁻²⁹ If a patient possesses some of these risk factors, the surgeon may dictate that rehabilitation to restore ROM progress at an accelerated pace relative to the normal protocol. These patient modifiers are very important in allowing for a successful postoperative rehabilitation.

MAINTENANCE AND RESTORATION OF MOTION

One goal of rehabilitation following surgery will be to restore normal motion at a rate that is dictated by the surgical procedure and what structures need to be protected.

Range of Motion

ROM is a joint's ability to achieve excursion in any given direction. ROM can be either passive or active. Passive ROM should be attained before the patient is able to achieve active control over motion, and thus passive ROM goals should come prior to active ROM goals in criterion-based protocols. In addition, passive ROM may be indicated prior to beginning active ROM in cases where muscle/tendon/bone interfaces require time to heal and muscle activation would be detrimental to healing. One example of such is the protected phase following rotator cuff repair where passive ROM is allowed prior to active ROM. Depending on involved tissue, passive and/or active ROM in specific directions may be indicated prior to ROM in other directions to protect healing tissue. For example, a subscapularis repair would have

restrictions on passive external rotation to limit overstretching, as well as restrictions on active internal rotation to limit stress from muscle activation.

Patients can experience decreased ROM for several reasons, and rehabilitation providers work to determine the cause of that restriction, as well as interventions to address it so that functional ROM can be obtained. Some reasons for decreased ROM are decreased flexibility and decreased joint mobility.

Immobilization

There are cases in which immobilization for a period of time is recommended to allow healing from surgical procedure. This would include situations with tenuous repairs, such as an Achilles repair, when the surgeon requires the patient to be casted in plantarflexion to allow protection and healing. A surgeon, with knowledge of the structures he or she wishes to protect, will dictate the period of time, as well as the position of immobilization. This is balanced with the concerns for stiffness. However, it should be noted that for some surgeries, such as rotator cuff repair, stiffness is transient and resolves within 6 months to 1 year following surgery.^{30,31} In contrast, stiffness following total knee arthroplasty (TKA) often requires manipulation or further surgery.³² Therefore the risk of revision surgery, failed tissue repair, and stiffness must be weighed when choosing immobilization strategies.

When to Begin “Passive” and “Active” Motion

Based on the balance in restoring ROM and protection, timelines for initiation of passive and active ROM are established. In cases that require protection of contractile tissue, patients will generally begin passive ROM prior to beginning active ROM. Passive ROM is the ability to move through an ROM without to use of contractile tissue (such as a rehabilitation provider moving a shoulder into external rotation while the patient is relaxed). Passive motion is intended to protect the repair interface of the contractile tissue, no matter which portion of the muscle-tendon-bone. It is important to note that no passive motion is completely passive without some muscle contraction. Most “passive” exercises exhibit 10% to 20% of a muscles maximum contraction during these exercises.³³⁻³⁵ In addition, when the joint is lengthened opposite the direction of the tissues under protection, the passive elements of the muscle will increase tensile forces. For this reason, passive ROM opposite the repaired muscle unit's action is generally limited for a period of time and progressed per surgeon direction. Active ROM generally follows passive ROM in these situations, particularly active in the direction of the muscle's line of action. Active ROM for a period of time based on healing properties of the repaired tissue.

In cases of cartilage lesions, the surgeon may restrict the patient from engaging these surfaces for a period of time. This may influence not only weight bearing but the portion of the joint's available motion that can be used. For example, following trochlear microfracture, knee flexion angle may be limited for a period of time based on the location of the lesion. The same is generally true for meniscal repairs, typically limiting end range flexion for 4 to 6 weeks. However, there is some evidence that continuous passive motion aids in healing cartilage and recovery

following cartilage procedures.^{20,36} This concept is based on regular joint motion stimulating synovial fluid production and bathing the tissues with the required micronutrients and lubrication.²¹ Our experience suggests regular, easy motion (continuous passive motion, stationary bike) provides good relief from stiffness and promotes the joint towards a “steady state” or normal following injury or surgery.

Joint Mobility

Joint mobility is the capacity of a joint to move prior to restriction from the soft tissues surrounding it, including the bony articulation, capsuloligamentous structures, and muscle/tendon structures. Often following surgery, due to immobilization, limited ROM, or an inflammatory response, patients can experience a decrease in their joint mobility that limits their available ROM. This is often related to a decrease in capsular volume or mobility, and rehabilitation providers will use techniques to improve the joint mobility so that functional ROM may be attained.

Flexibility

Flexibility is a muscle's ability to lengthen to allow normal motion. Impairments in flexibility are commonly seen after surgical procedures, even those which do not have muscular involvement. Immobilization or limited ROM about a joint for protection following surgery often dictates that muscles surrounding the joint are not able to be stretched through their full excursion, and over time, these muscles lose flexibility. As ROM is allowed, rehabilitation providers will also work to restore normal flexibility of surrounding musculature. If decreased flexibility is the limiting factors in restoring ROM, rehabilitation providers will develop focused programs to improve flexibility through stretching and manual therapy techniques.

RESTORATION OF MUSCLE PERFORMANCE

Another goal following surgery will be to restore muscle performance at the rate that allows for congruency and healing of tissue structures.

Inhibition/Activation

Some muscle groups tend to be inhibited following surgery or injury, which can result in aberrant activation of muscle groups about a joint. One such example is the quadriceps, which tends to be inhibited following knee trauma or surgery due to pain and swelling, as well as spinal and cortical level changes.^{37,38} Following most knee surgeries, a primary goal in early postoperative rehabilitation is to improve the activation of the quadriceps. Examples of techniques are neuromuscular electrical stimulation (NMES), biofeedback, transcranial stimulation as well as careful monitoring of swelling and pain at the knee.³⁹⁻⁴²

Hypertrophy Versus Motor Learning

One large component of postoperative rehabilitation is strengthening. There are many different ways to consider strength, which are outlined later in “Muscle Performance.” When we consider strengthening in postoperative rehabilitation, it is important to

note that early changes are largely due to improvement in the efficiency and effectiveness of motor unit firing.^{43,44} It takes much longer (at least 6 to 8 weeks) to see changes due to muscle hypertrophy, or growth of the size of the muscle. Early postoperative rehabilitation should focus motor learning and improving the ability of muscle groups to fire appropriately, where later rehabilitation should focus on hypertrophy, or increasing the muscle's size.^{45,46}

PRINCIPLES OF CRITERION BASED PROGRESSION

Progression in rehabilitation should be both time and criterion based. *Time-based criteria* are based on setting anatomic stoplights and provide the guideposts for rehabilitation programs. These timelines are aimed to protect tissue from loading that could disrupt or slow healing following surgery. These timelines are tissue specific and are limited because they are only estimates based mostly on animal studies and normal variation in individual patient healing. *Criterion-based progression* requires specific criteria to be achieved to move forward to the next phase of rehabilitation. These criteria are aimed at ensuring specific impairments are resolved along the recovery path. This optimizes the patient's ability to match their activities to allow a safe progression back to their desired level of functional. Therefore the time-based criterion guide the fastest that a patient could safely progress through rehabilitation, whereas the criterion-based progression may require longer and is patient specific.

Key Criteria for Rehabilitation Protocols and Progressions

Outcome Measures

There are many patient-reported outcome (PRO) measures which are used to assess patients' perception on some combination of their pain, function, and satisfaction. These measures can be specific to a joint, such as the PENN Shoulder Score⁴⁷ or International Knee Disability Committee (IKDC).⁴⁸ They can also be broad to capture a patient's overall health views, such as the Veterans Rand 12-item Health Survey (VR-12).⁴⁸ The American Academy of Orthopaedic Surgeons Quality Outcomes Data (QOD) Work Group on Patient Reported Outcome Measures has recommended specific PROs for general health and each body region.^{49,50} These PROs were recommended based on the following criteria⁵¹:

1. PROs
2. Validated scores
3. Good psychometrics
4. Ease of use for the patient (i.e., brief)
5. Ease of scoring and understanding for physician
6. Standardized use nationally
7. Consideration for cost

Specifically, selected PROs should demonstrate good psychometrics including reliability, validity, and minimal clinically important differences (MCIDs) for the population in question. Individual patient progression compared with population norms at regular intervals throughout the recovery process can aid discharge planning and long-term prognosis.^{51,52}

Pain

Postoperative pain is expected and normal in association with many procedures. Adequate pain control is requisite for progression through rehabilitation. If pain is not controlled, progression to more demanding activities will likely exacerbate pain. As pain is known to have an inhibitory effect on certain muscle groups, pain control is also essential for proper muscle performance and progression through muscle performance activities. Pain levels, generally through a Numeric Pain Rating Scale (NPRS) or visual analog scale (VAS) should be assessed regularly and monitored, and an increase in pain may signal a need to decrease demands of rehabilitation or modify activities accordingly. In general, minimal to no pain should be experienced both within and between rehabilitation sessions prior to progressing a patient to the next phase of rehabilitation. There are some surgical procedures and postoperative rehabilitation protocols which can be associated with increased pain, and patients should be educated accordingly for matched expectations of their progress.

Range of Motion

ROM can be measured passively as well as actively. In general, passive ROM should be attained before active ROM, as dictated by protecting healing tissue. Time-based parameters should be established for when passive and active ROM are allowed and to what extent. Criteria should be established for what amount of passive ROM and active ROM are required prior to progressing within a protocol.

Muscle Performance

There are many different measures of muscle performance. *Strength* refers to the maximal force output of a muscle group to a load. It is commonly measured in the physical therapy setting through the use of a dynamometer for a quantifiable measure or through the use of manual muscle testing when dynamometers are not available. Strength should be a component of any criterion-based protocol, with discharge criteria generally being set as attaining at least 90% strength relative to the contralateral limb.⁵³⁻⁵⁵ Many protocols also cite a balance of strength between muscle groups, such as external rotation/internal rotation strength ratio of the shoulder in throwers, or quadriceps to hamstrings strength ratio on lower extremity athletes.^{56,57} *Power* is defined as a muscle group's ability to produce work over a period of time and is often termed "rate of force development" in recent literature.^{58,59} Power is proportional to the speed at which one can produce a force. There are many different ways to measure power, and in general, these tests are performed in patients whose activity level demand explosive movements, such as those who perform sprinting or jumping. *Endurance* is another measure of muscle performance, and it relates to a muscle group's ability to maintain strength over a period of time while under load.⁶⁰ Patients may use compensatory muscular or movement patterns when fatigued that increase their susceptibility to subsequent injury. There are many ways to measure endurance and fatigue, and these measures relate to placing patients in the fatiguing activity they would be performing, to assess their ability to

sustain a level of performance and how they are able to perform when fatigued.

Balance

Balance refers to one's ability to maintain their center of mass over their base of support without losing stability. Balance is a common impairment seen in patients following lower extremity surgery, especially amongst those who have compromised tissue that would otherwise contribute to congruency or proprioception. There are many ways to measure balance. In general, a measure of balance is commonly cited in criterion-based protocols in order to progress from assistive devices with gait activities. A measure of balance should also be cited prior to discharge to ensure adequate stability for a patient's full return to function. The Star Excursion Balance Test (SEBT) and modified SEBT are commonly used (Fig. 31.4).^{61,62}

Movement Quality

Movement quality refers to a patient's ability to perform movements against a normative standard of movement. Deficits in movement quality, including "faulty movement patterns," have been identified as risk factors for future injury. Movement quality is assessed throughout rehabilitation, with movement retraining integrated as necessary in care. Examples of movement quality tests include scapula dyskinesis during arm elevation, the Landing Error Scoring System (LESS) for jumping landing mechanics, or the Functional Movement Screen for squatting.⁶³⁻⁶⁶



Fig. 31.4 The anterior direction of the modified Star Excursion Balance Test.

Return-to-Sport Considerations

The decision regarding when to return an athlete to sport is complex. There should be both time and criterion-based considerations based on both anatomic healing and functional performance. This allows the athlete to return at a point which optimizes the balance between minimizing time loss and maximizing function and ability to return safely without subsequent injury. This requires a team effort because the surgeon is best equipped to decide if the tissues have appropriately healed and the rehabilitation provider can assess the athlete's ability to function and the constructs associated with successful return to sport. It is the responsibility of the rehabilitation providers to communicate their assessment of the athlete's functional capabilities and readiness to return to sport to the athlete's surgeon. Recent evidence shows that those who return too soon (<7 months following ACL reconstruction) or do not meet the return-to-sport criteria are at significant risk for subsequent injury.

In addition to the factors referenced previously (pain, ROM, muscle performance, and outcome measures), there are additional functional measures that should be assessed when determining an athlete's readiness to return to sport.⁶⁷⁻⁷¹

KEYS TO SUCCESSFUL REHABILITATION

Role of Preoperative Rehabilitation

Preoperative rehabilitation should be included as a portion of a patient's comprehensive surgical care. In our experience, preoperative rehabilitation has many benefits.⁷²⁻⁸¹

- A preoperative visit allows a rehabilitation provide to establish baseline values for a patient's pain, ROM, and strength, along with other measures deemed appropriate.
- It allows patients to learn more about their postoperative rehabilitation process prior to surgery, when they are not dealing with postoperative pain and impaired cognition from postoperative medications. Patients learn what they can expect to feel and do during each phase of their postoperative case, as well as an estimated timeline on their duration of care to reach their functional goals. Having this conversation prior to surgery allows the patient time to plan accordingly so that they go into surgery with clear postoperative expectations for rehabilitation.
- It allows the rehabilitation provider an opportunity to provide recommendations that would be helpful prior to surgery to decrease pain and swelling, restore ROM, and restore muscle performance as appropriate in a safe manner. Many studies illustrate a relationship because preoperative function and postoperative function, so this allows patients to optimize their preoperative function so that they may have a more successful postoperative outcome.
- It allows patients an opportunity to receive training and education on any postoperative equipment they will be using, including assistive devices and protective devices such as slings or braces. This allows them the opportunity to practice prior to surgery.

Communication

Surgical protocols and referrals are used to detail the expected postoperative progression based on the surgery performed. These protocols are guidelines used by rehabilitation providers to progress patients accordingly. They may differ from surgeon to surgeon, but in general, their key is their ability to match dosing of load to allow the patient to heal, progress safely, and return to their activities with a minimized risk of future injury. The protocols combined with a detailed referral are the keystones of matching expectations. The surgeon should provide a detailed referral to guide the rehabilitation provider to deliver the care expected by the surgeon. The referral should at minimum include:

- *Diagnosis:* Postoperative referrals should detail the patient diagnosis, including what tissue damage was identified preoperatively and intraoperatively.
- *Operative details:* The postoperative referrals should outline any surgical procedures performed, including location and size of repairs or removal of tissue.
- *Patient modifiers:* As discussed regarding patient modifiers, a surgeon may wish to alter an established protocol based on patient-specific factors, whether specific tissue factors in the surgery, such as poor bone quality in a rotator cuff repair, or patient-specific factors, such as a metabolic concern for altered healing response. These modifications should be outlined on a postoperative referral if they deviate from the established protocol. If the surgeon wishes to modify a protocol based on patient presentation at a time point following surgery after a follow-up visit, this should be communicated to the patient's rehabilitation provider as well.

Matching Expectations

From the diagnosis through return to sport or activity, setting expectations for a patient will aid in smoothing through the difficult process. This begins with the preoperative visit and outlines for the patient the general timeline in which they can expect to progress following surgery. This way, patients can plan for expected time to use assistive devices or braces and to return to activities, including, driving, work, sports, etc. The protocol then establishes when progression is safe and expected for a standard surgical procedure. This allows rehabilitation providers and surgeons to be consistent in their message delivered to patients to minimize any confusion and create a positive patient experience.

Timing of Communication and Follow-Up

Surgeons vary in the amount and frequency of communication they expect from rehabilitation providers. In general, rehabilitation providers should communicate patient progress to surgeons prior to each of the patient's appointments with the surgeon. They should also communicate with the surgeon if they have concerns, including a new injury or failure to progress as expected, even if the patient does not have an upcoming appointment with the surgeon in the near future. They should never progress faster than a protocol timeline allows, without communication from the surgeon to do so. Following these principles will optimize communication and ultimate outcome for patients following surgery.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

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